

FX Models for Quants

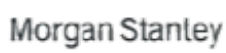
Pricing, Hedging, Smile & PnL Across Currency Products
Why Desks Choose GK, Heston, LSV, Jump-Diffusion — and When They Break

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Model	Core Assumption	State Variables	Smile Dynamics	Jump Handling	Calibration	Products	Primary Failure Mode	Desk Usage
Garman-Kohlhagen	Lognormal, constant vol	Spot S_t	Static	Ignored	ATM vol	Vanilla Europeans	No smile, gaps	Quoting standard
Local Vol (Dupire)	$\sigma(S, t)$ deterministic	Spot S_t	Sticky strike only	Ignored	Full surface	Barriers, digitals	Wrong forward smile	Legacy exotics
Heston	Mean-reverting variance v_t	S_t, v_t	Sticky delta	Vol clustering only	ATM + skew	Vanilla books, variance swaps	No jumps, short-date skew weak	Risk management
SABR	Stoch vol + CEV	F_t, α_t	Flexible (β)	Wings only (asymptotic)	RR, BF, wings	Swaptions, vanillas	Deep OTM failures	FX vanilla standard
Jump-Diffusion	Poisson jumps + diffusion	S_t, N_t	Jump size distribution	Explicit jumps	OTM put premium	Short-dates, EM FX	Jump frequency unstable	Earnings/weekends
Hybrid FX-Rates	Correlated rates + FX	S_t, r_d, r_f	Rate vol feeds FX vol	Ignored	FX + rates vols	Quanto options, dual-currency notes	Correlation explosion	Structured products

Table 1: FX Models Landscape: Product → Risk → Model Mapping

Table 2: Product Complexity Ladder - When to Upgrade Your Model

Level	Product Class	Model Choice	Why Upgrade?	Calibration Time
1	Vanilla European (ATM, near-ATM)	GK or SABR ($\beta = 1$)	Fast quoting, clean Greeks	<1 second
2	Vanilla with skew (25Δ, 10Δ)	SABR (full params)	Need RR/BF fit	1-5 seconds
3	Single barriers, touch options	LSV or SABR-LMM	Sticky-delta dynamics matter	10-30 seconds
4	Double barriers, window barriers	LSV + MC simulation	Path-dependent + surface movement	1-5 minutes
5	EM short-dated, policy-sensitive	Jump-Diffusion (Bates/Merton)	Gaps break continuous hedging	5-10 seconds
6	Quanto options, dual-currency notes	Hybrid FX-Rates (3-factor)	Correlation is first-order risk	30-60 seconds
7	Multi-asset worst-of, baskets	Multi-SABR + Copula	Cross-currency correlation	2-10 minutes
8	PRDCs, exotic autocallables	Custom LSV + Jump + Correlation	Multiple risk factors interact	10-60 minutes

Desk Reality Check

The Speed-Accuracy Trade-off:

- **Real-time quoting (Levels 1-2):** Must price in <5 seconds → SABR only
- **Client trade (Levels 3-4):** Can take 1-5 minutes → LSV acceptable
- **Monthly reserves (Levels 5-8):** Can take hours → Full MC simulation

Interview Gold:

"I choose the *simplest* model that captures the *dominant* risk. Overengineering wastes time and introduces parameter instability. A well-hedged SABR position beats a poorly-calibrated LSV position."

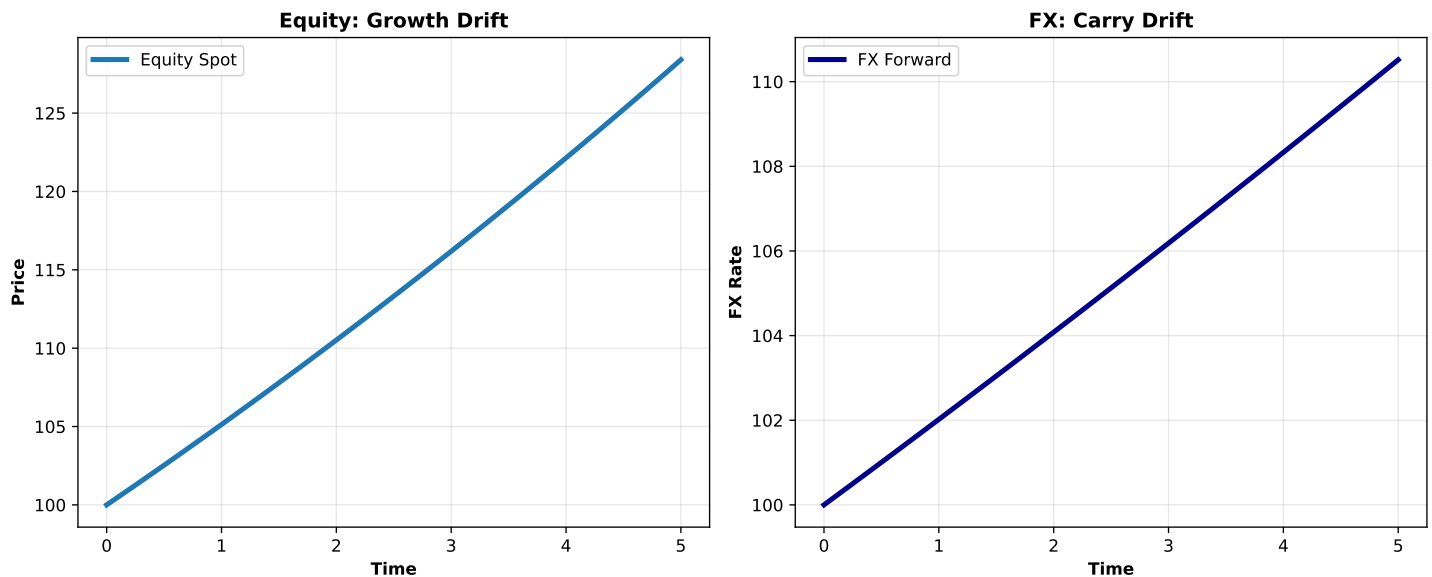


Figure 3: Forward vs spot drift: Equity grows exponentially; FX forward is shaped only by carry differential.

Practitioner Insight

The GK Paradox:

Every desk knows GK is wrong (no smile, no jumps), yet:

- 90% of FX options are quoted in "GK vol"
- Risk systems use GK delta for position limits
- Regulators require GK vega for capital

Why?

1. **Common language:** Everyone understands "15 vol" means GK vol
2. **Speed:** Closed-form pricing in microseconds
3. **Stability:** GK greeks don't jump around daily
4. **Regulatory:** Basel requires standardized approach

The Trick: Use GK for *quoting and hedging*, but use SABR/LSV for *pricing and reserves*.

Interview Line: "GK is the coordinate system. Just like we use Cartesian coordinates even though spacetime is curved — you need a reference frame everyone agrees on."

Module 8: Hedging & PnL in FX—Carry Matters

Module Overview

What this module is about: This module connects FX models to real desk PnL by decomposing profits and losses into spot, carry, volatility, and residual components.

Why this matters: A model is only useful if it explains where PnL comes from and why hedges succeed or fail.

What confusion this clears:

- Why carry dominates FX PnL
- Why vega-neutral books still lose money
- How residual PnL signals model failure

After this module, you should be able to answer: “How do FX desks explain daily PnL and detect model breakdowns?”

8.1 FX PnL Decomposition

Total PnL:

Core Formula

$$\text{PnL} = \underbrace{\Delta_{\text{spot}} \times \Delta X}_{\text{Spot Delta}} + \underbrace{(r_d - r_f) \times X \times \Delta t}_{\text{Carry}} + \underbrace{\mathcal{V} \times \Delta \sigma}_{\text{Vega}} + \underbrace{\mathcal{S} \times \Delta \text{RR}}_{\text{Skew}} + \text{Residual}$$

Carry PnL: Unique to FX, dominates for forwards **Vega PnL:** From vol moves **Skew PnL:** From RR changes

"FX PnL = Delta + Carry + Vega + Skew"

Memory: Carry is 30-50% of PnL for forward-dated options. **Interview:** "I attribute FX PnL by carry first, then delta/vega."

8.2 Carry Hedging Reality

Problem: Carry is deterministic but FX spot is stochastic **Solution:** Hedge forward delta, not spot delta

Core Formula

$$\Delta_{\text{forward}} = \Delta_{\text{spot}} \times e^{r_f T}$$

FX Carry Hedge

"Forward delta hedges both spot and carry"

"Spot delta alone misses carry term"

"Interview: 'I hedge forwards with forward delta, not spot delta.'"

8.3 Model Risk vs Market Risk

Model risk: Residual PnL from wrong dynamics assumptions (sticky strike vs delta)

Market risk: Actual spot/vol moves

Detection: - Residual > 5% daily PnL → model broken - Residual spikes on policy dates → jump model missing